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PROJECT PROGRESS REPORT

DESIGN OF A LEO CUBESAT CONSTELLATION FOR REMOTE MONITORING

A **Final-year Project Progress Report** submitted in partial fulfilment
of the requirements for the degree of
Bachelor of Engineering

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ABSTRACT

This progress report presents the design of a Low Earth Orbit (LEO) CubeSat constellation for remote monitoring of Nepal's glacial lakes, developed in MATLAB using the Satellite Communications Toolbox. Rather than fixing the orbit configuration by convention, e.g. a Sun-synchronous or Walker-Delta pattern chosen in advance, this work poses orbit selection itself as a black-box optimization problem solved with Bayesian Optimization which consists of steps: Latin Hypercube sampling, Gaussian-Process surrogate, and an Expected-Improvement acquisition function. First to find the best possible single-satellite orbit and then to grow the constellation one satellite, or one orbital plane, at a time.

Each growth step is scored by a composite objective that combines a weighted mission score: coverage fraction, revisit time, ground-station contact time, downlink latency and per-lake coverage with a Wasserstein or earth-mover distance term that rewards a candidate satellite whose daily pass-time distribution is temporally complementary to the existing constellation's, plus explicit largest-gap and gap-uniformity terms that directly target a narrow, evenly spread daily revisit schedule.

Keywords: *CubeSat, LEO satellite, satellite constellation, Bayesian Optimization, Wasserstein distance, incremental constellation growth, remote monitoring, Nepal, GLOF, revisit time*

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SYMBOLS AND ABBREVIATIONS

ADCS	Attitude Determination and Control System
BO	Bayesian Optimization
EI	Expected Improvement
GDOP	Geometric Dilution of Precision
GLOF	Glacial Lake Outburst Flood
GP	Gaussian Process
LEO	Low Earth Orbit
LHS	Latin Hypercube Sampling
MATLAB	Matrix Laboratory
PPP	Precise Point Positioning
RAAN	Right Ascension of the Ascending Node
SAR	Synthetic Aperture Radar
SGS	Satellite Ground Station
SGP4	Simplified General Perturbations (4th order) orbit-propagation model
TLE	Two-Line Element set
e	Eccentricity
R (or a)	Semi-major axis
i	Inclination
Ω	Right ascension of the ascending node (RAAN)
ν (θ)	True anomaly
Δh	Altitude offset

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CHAPTER I: INTRODUCTION

1.1 Background and Description

Advances in digital technologies now let Earth observation happen without anyone physically present at the site. A CubeSat carrying a small imaging or radio-frequency sensor can monitor a remote or inaccessible region continuously and a coordinated constellation of CubeSats can cover a country with frequent passes, day and night, almost regardless of weather, because Low Earth Orbit (LEO) gives short orbital periods and therefore fast revisit times. This project designs such a constellation for Nepal, where mountainous terrain makes ground-based hazard monitoring difficult and expensive to sustain.

What distinguishes this project from a conventional CubeSat constellation study is how the orbit is chosen. Rather than adopting a standard pattern such as a Sun-synchronous or Walker-Delta configuration outright, the project treats orbital-parameter selection itself as an optimization problem: a MATLAB pipeline built on the Satellite Communications Toolbox propagates candidate orbits (inclination, Right Ascension of the Ascending Node (RAAN), true anomaly, and altitude offset), simulates the resulting coverage of Nepal over a 24-hour scenario and scores each candidate on a weighted combination of coverage fraction, revisit time, ground-station contact time, downlink latency and coverage of six named glacial lakes. A single best orbit is found first by Bayesian Optimization; the constellation is then grown one satellite or one new orbital plane, at a time, with every growth step itself chosen by a further Bayesian Optimization search.

1.2 Problem Definition and Motivation

Nepal's mountainous terrain hosts glacial lakes, forests, and river valleys prone to Glacial Lake Outburst Floods (GLOFs), landslides, forest fires and general flooding. Timely alerts require observing these remote areas frequently, but many of the most endangered regions lack ground sensors or permanent infrastructure. A LEO CubeSat network could fill that gap by imaging these areas from space, provided its orbit configuration (number of satellites, number of orbital planes, inclination, phasing) actually keeps every part of Nepal in view often enough, in both daylight and darkness, to detect a hazard quickly.

A constellation can have an excellent average revisit time and still leave a multi-hour block of the day with no coverage at all, simply because its passes happen to cluster together. This project's

central technical problem is to search the very large space of possible orbital parameters efficiently enough that a full-fidelity propagation and coverage simulation can be the objective function rather than a cheap analytic approximation.

1.3 Objectives

The objectives of the project are:

- To formulate LEO orbital parameters as a Bayesian Optimization problem for single-satellite and incrementally grown multi-satellite, multi-plane constellations, maximizing Nepal coverage while minimizing revisit time for six at-risk glacial lakes.
- To develop temporal-complementarity criteria using Wasserstein distance, largest-gap, and gap-uniformity measures to promote narrow and evenly distributed daily revisit intervals.
- To verify coverage and revisit performance through independent geometric checks and per-lake analysis, establishing day-and-night monitoring capability for Nepal's highest-risk glacial lakes.

1.4 Significance of Study

Establishing a dedicated CubeSat constellation for Nepal would meaningfully improve disaster response: constant monitoring through multiple, well-phased satellites reduces revisit time over critical zones and even a small constellation can enable near-real-time situational awareness in places a ground network cannot reach. This is particularly significant for a developing country like Nepal, which has suffered devastating floods and GLOFs and cannot easily fund a dense ground-sensor network.

Technically, the project is also a case study in optimization-driven constellation design. Most published regional-constellation studies fix a constellation pattern, typically a Walker configuration and then search within it, or use a genetic algorithm to search the full parameter space. This project instead applies sample-efficient Bayesian Optimization directly to a full-fidelity coverage simulation and extends the objective itself with a Wasserstein-distance temporal-complementarity term and explicit gap/uniformity criteria so that the search targets the actual shape of the daily revisit schedule, not only its mean. Bayesian Optimization driving a physically simulated, multi-objective, temporally aware growth loop, is the project's main technical contribution.

CHAPTER II: LITERATURE REVIEW

This section explains the relevant studies which have been already conducted relating to the small satellites based low orbit constellation design in different domains focusing on disaster management applications. Here, three representative studies that inform required design are discussed.

Sayanju et al. (2022) describe Nepal's Danfe Space Mission, a 3U CubeSat demonstration for glacial lake outburst flood monitoring. Danfe integrates a LoRa-based uplink from ground sensors measuring lake water level. The authors argue that existing early-warning systems are ground-cellular and unreliable in remote mountains, so a "ground-to-space monitoring system" is needed. The 3U Danfe CubeSat uses a lightweight open-source platform running a drone OS on an STM32 SoC to relay sensor data. They conclude that if Danfe works, future constellations could be built much faster and cheaper, providing near-real-time water-level data to prevent floods. This study is directly relevant: it demonstrates CubeSat technology and LoRa communications in Nepal, and explicitly envisions scaling to a constellation [1].

Santilli et al. (2018) present a CubeSat constellation for disaster monitoring in Brazil. They note that "real-time, fast and continuous information broadcast is a fundamental requirement" for disaster management, and that a CubeSat network "can considerably decrease the revisit time" by deploying multiple satellites in orbit. Their design uses a Walker-type constellation: they calculate the number of satellites and orbital planes needed to minimize revisit time over a target region. Each CubeSat carries appropriate sensors and has autonomy in pointing and control to meet mission needs. This work shows how a properly distributed small-satellite network can monitor remote areas such as flood-prone basins effectively [2].

Rashed and Bang (2022) focus on optimizing small-satellite constellations for disaster management. They observe that current commercial constellations are expensive and inaccessible for many regions, and propose design techniques to optimize coverage and cost. Their simulations show that an optimized LEO constellation can relay data faster and with precision while significantly reducing costs. They emphasize autonomous orbit selection and trajectory optimization so that satellites maximize coverage and minimize revisit time. Although their work is broad and also considers deep-space support, the key takeaway is that strategic constellation

design through choice of inclinations, spacing, and autonomy can meet disaster-management needs in a cost-effective way [3].

Tim Schubert (2022) introduced ns-3-leo, an extension of the discrete-event network simulator NS-3 specifically designed for LEO satellite networks. This module models satellite mobility, link characteristics, and protocol behavior, enabling the evaluation of existing routing protocols in high-mobility environments. Their work demonstrated that conventional flooding-based approaches are insufficient for broadband connectivity in LEO constellations, highlighting the unique challenges of handover, Doppler shifts, and dynamic coverage, ns-3-leo provides a validated framework to simulate communication link performance, handover behavior, and network reliability, thereby informing constellation design and enabling efficient data relay between satellites and ground stations [4].

He et al. (2020) addresses the challenge that natural disasters occur unpredictably and traditional fixed constellations can only observe a disaster site when a satellite happens to pass overhead. The authors design a reconfigurable LEO satellite constellation in which satellites can be commanded to transfer orbits and fly directly over a disaster site on demand. Using a physical programming approach combined with a genetic algorithm, they co-optimize multiple competing objectives, minimizing velocity increment for orbital transfer, minimizing maximum and mean reconfiguration time, and reducing variability in reconfiguration time across all possible disaster scenarios. Their results demonstrate that a well-designed reconfigurable constellation achieves far superior observation performance over a disaster area compared to static designs, while remaining fuel-efficient. This work is directly relevant because it establishes a rigorous mathematical framework for constellation design that balances coverage, responsiveness, and cost for disaster monitoring, the same trade-space central to any LEO remote monitoring mission [5].

Sah, Srivastava, and Das (2021) develop a LEO small satellite constellation specifically for continuous infrastructure monitoring over a defined regional footprint (India), using a Synthetic Aperture Radar (SAR) payload. They design 4 orbital planes at 600 km altitude with an inclination of 36 degrees, each plane containing 3 satellites separated by 120-degree true anomaly spacing, for a total of 12 satellites. The authors select S-band for telemetry, tracking and command while using X-band and VHF for data downlink, and they equip each satellite with Control Moment Gyros for high-slew-rate pointing. Their coverage analysis shows that the average revisit time

ranges from 15-35 minutes, well under one hour, with near-100% coverage efficiency over the target region. The study is highly relevant because it provides a concrete, end-to-end methodology for designing a regional LEO constellation with defined coverage metrics, applicable directly to any geographically constrained remote monitoring mission [6].

Azami et al. (2022) present the KITSUNE 6U CubeSat mission, a collaboration between Kyushu Institute of Technology, Harada Seiki, and Addnics Corp., deployed from the International Space Station in March 2022. The platform carries a 31.4-megapixel commercial off-the-shelf sensor with customised optics achieving 5-metre ground sampling distance, demonstrating that CubeSat-class hardware can deliver high-resolution Earth observation from LEO. As a secondary mission, the authors implement and validate a Convolutional Neural Network (CNN) onboard the satellite for in-orbit wildfire image classification, achieving 97-98% accuracy using the MiniVGGNet model. Processing images onboard rather than at the ground station substantially reduces downlink data volume. The study is relevant because it proves that modern 6U CubeSat platforms are capable of high-resolution remote sensing with onboard AI processing, a key capability for any disaster monitoring constellation seeking to relay actionable intelligence rather than raw imagery [7].

Imoto, Satoh, Obata, and Yamada (2023) propose an optimisation methodology for Earth observation constellation design based on the geometry of satellite ground tracks, applied specifically to the requirements of Interferometric SAR (InSAR) missions. The authors mathematically formulate orbit conditions required for InSAR coherence, then determine the nadir trajectory configurations, the ground track layouts that satisfy coverage requirements with the fewest satellites. Using multi-objective optimisation, they derive a Pareto-optimal set of constellation designs that trade off the number of satellites against revisit performance for various mission conditions. Unlike conventional Walker-based designs that impose uniform inclination and spacing, their ground-track-centric approach adapts to mission-specific coverage needs. This work is directly relevant because it provides a rigorous, mission-driven methodology for minimising satellite count while guaranteeing coverage, a principle applicable to any LEO remote sensing constellation with constrained budgets [8].

Wu et al. (2020) present the STU-2 (TW-1) mission, a three-satellite CubeSat constellation developed at Shanghai Jiao Tong University consisting of one 3U and two 2U CubeSats launched together on 25 September 2015 into a LEO sun-synchronous orbit, forming an along-track

formation. Each satellite carries a different payload, enabling the constellation to simultaneously demonstrate integrated Earth observation and marine/air traffic monitoring using Automatic Identification System (AIS) and Automatic Dependent Surveillance-Broadcast (ADS-B) receivers. The paper reports in-orbit test results and operational lessons learned from the distributed payload architecture, demonstrating that a small multi-unit constellation with heterogeneous instruments can provide complementary data streams cost-effectively. The study is relevant because it is one of the few papers to report real operational results from a CubeSat constellation carrying diverse remote monitoring payloads, validating the distributed-satellite approach at low cost and providing lessons directly applicable to multi-sensor constellation design [9].

Jiang, Luo, Guan, and Zhu (2024) design a low-orbit regional enhanced navigation constellation for China's BeiDou-3 (BDS3) system using a Bayesian Optimization algorithm - the same optimization framework this project applies to Nepal coverage. The authors note that constellation configuration parameters cannot be related to navigation performance through a closed-form model, so the design problem is treated as a black-box optimization: a Gaussian-Process surrogate models Geometric Dilution of Precision (GDOP) and the average number of visible satellites as functions of orbital inclination, RAAN, and mean anomaly, while an Expected-Improvement acquisition function selects each successive candidate constellation. Four Bayesian-Optimization-designed LEO configurations (70-81 satellites at 1000 km altitude) are shown to reduce BDS3's global GDOP by 18-20% and shorten Precise Point Positioning convergence time by more than 80% within the targeted (Chinese) region, while converging roughly 54% faster than an equivalent genetic-algorithm search. This study is directly relevant: it is one of the few papers to apply the same Gaussian-Process / Expected-Improvement Bayesian Optimization framework used in this project to a regional satellite-constellation design problem, and it demonstrates, in a navigation context, the same kind of region-specific performance gain - and the same efficiency advantage over a genetic-algorithm baseline - that this project targets for Nepal's disaster monitoring [10].

The surveyed works confirm that a distributed LEO system of small satellites is a viable solution for monitoring remote disasters. They highlight the importance of minimizing revisit time by having multiple satellites in well-chosen orbits.

CHAPTER III: RESEARCH METHODOLOGY

3.1 Orbital Design Simulation

The orbital design pipeline is implemented as a single MATLAB script built on the Satellite Communications Toolbox. It is organized into five stages, shown in Figure 1: (i) initializing the 24-hour scenario clock and loading Nepal's geometry (the national border, from a shapefile and six named glacial-lake targets plus a ground station); (ii) finding the best possible single-satellite orbit by Bayesian Optimization; (iii) growing that single satellite into a multi-satellite, multi-plane constellation, one Bayesian-Optimization-chosen step at a time; (iv) computing coverage, revisit, ground-station and per-lake metrics for the result; and (v) producing static and animated 2-D ground-track maps, a 3-D satellite scenario viewer, and a pass-time histogram for inspection and reporting. The subsections below describe stages (ii) to (v) in turn; this pipeline, and the Bayesian-Optimization-driven growth loop in particular, is the core of the project.

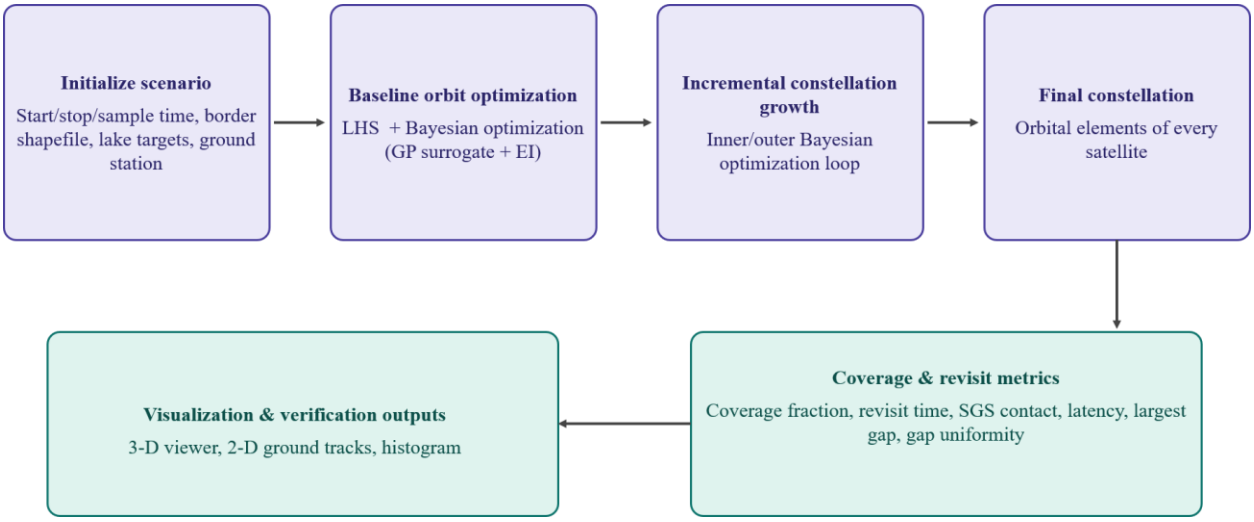


Figure 1: Orbital Design Simulation Architecture

3.1.1 Baseline Single-Satellite Orbit Optimization

The orbit of a single satellite is fully described, for this project's purposes, by its inclination, RAAN, and true anomaly. Altitude is held fixed for the baseline search. These three parameters are searched over bounds of inclination $\in [0°, 90°]$, RAAN $\in [0°, 360°]$ and true anomaly $\in [0°, 360°]$. Rather than evaluating this space on a grid, or assuming an analytic coverage model, the search treats the full propagate-and-simulate pipeline as an expensive black-box function and optimizes it with Bayesian Optimization: 100 initial candidates are drawn by Latin Hypercube

Sampling (LHS) so the early samples are well spread across the parameter space, each is scored by the same weighted mission score used throughout the project, a Gaussian-Process (GP) surrogate is fit to all scored candidates and an Expected-Improvement (EI) acquisition function proposes the next candidate most likely to improve on the best score found so far. This loop, shown in Figure 2, repeats for a further 100 iterations. The weighted mission score combines coverage fraction, average revisit time, ground-station (SGS) contact time, downlink latency, and glacial-lake coverage, each normalized by its own scale and combined with a hand-set weight, so that no single metric dominates the search by virtue of its units.

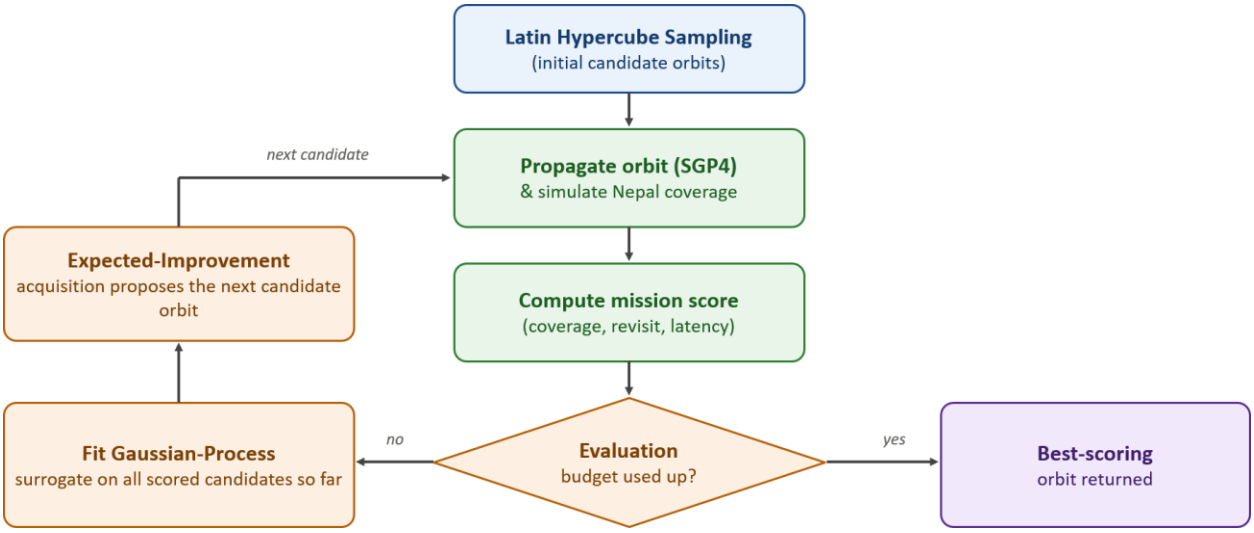


Figure 2: Bayesian Optimization Loop

3.1.2 Incremental Constellation Growth

Starting from the best single-satellite orbit, the constellation is grown one decision at a time. At each step, the loop shown in Figure 3 first asks whether the currently active orbital plane still has a free satellite slot, capped at 3 satellites per plane in this project: if so, an "inner" Bayesian Optimization searches true anomaly and an altitude offset: ±40 km, kept ≥ 15 km from any co-plane satellite to avoid a collision risk, to place a new satellite within that same plane. If the plane is full, or every inner candidate is vetoed, an "outer" Bayesian Optimization instead searches inclination and RAAN for an entirely new orbital plane, with a short nested search choosing that new plane's first satellite's true anomaly and altitude offset. Both searches use the same LHS-seeded, GP/EI Bayesian Optimization loop as the baseline search shown in Figure 2, but score each candidate with the extended growth objective described in 3.1.3 rather than the mission score alone.

A candidate is accepted once it clears any one of several gain thresholds, a minimum absolute improvement in coverage fraction, in average revisit time, in downlink latency or in the largest daily pass-time gap, and a new orbital plane, once found, is always adopted, since at that point in a step it is the only remaining growth action available. Both the inner and outer searches are subject to a hard veto first: any candidate that would let the worst-case (per-lake) revisit time regress by more than 15 minutes is excluded from selection outright, regardless of how well it scores otherwise; if every candidate in a step's pool trips that veto, the growth loop stops rather than accept a regression. The loop otherwise continues until a fixed budget is reached: at most six growth steps and three orbital planes, with at most three satellites per plane.

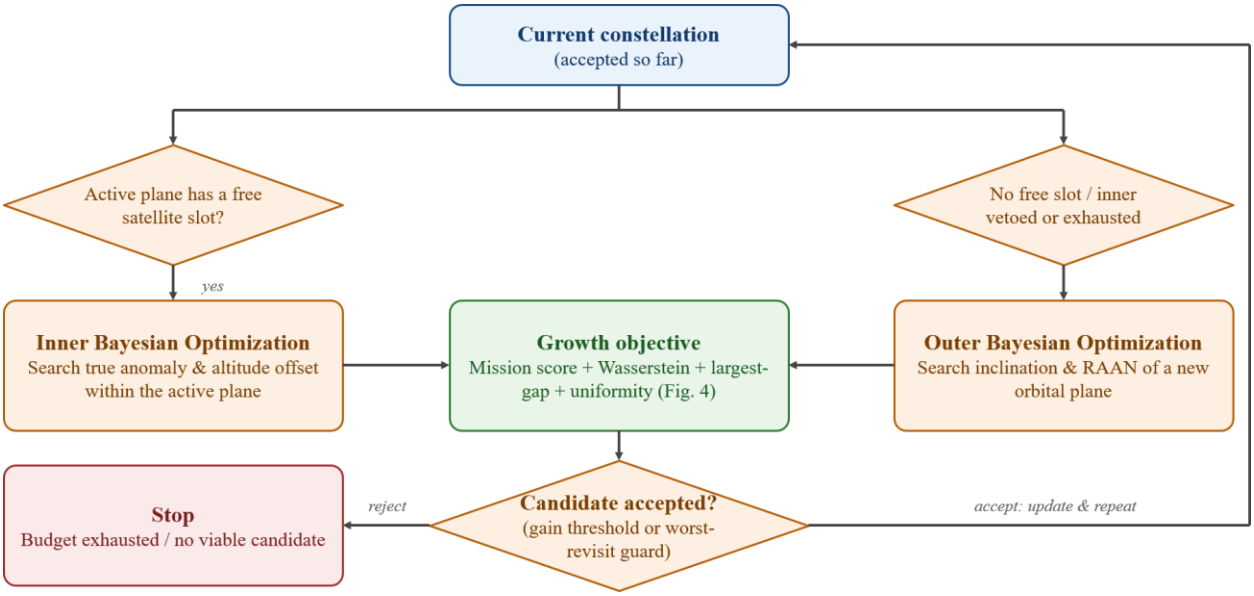


Figure 3: Incremental Constellation Growth Loop

3.1.3 Growth Objective

The growth objective that ranks every candidate satellite or orbital plane is a sum of four terms. The first is the same weighted mission score used for the baseline orbit: coverage, revisit, latency, lake coverage. The remaining three terms are specific to constellation growth, and are what let the search target the shape of the daily revisit schedule rather than only its average.

- Wasserstein (earth-mover) distance: For the existing constellation and for the candidate satellite alone, the daily pass-time-of-day distribution, a 24-hour histogram of when passes over Nepal occur, is computed, and the Wasserstein distance between the two distributions is added to the objective as shown in Figure 4. This distance is large when the candidate satellite passes occur at times when the existing constellation provides little coverage and small when

they occur at similar times. Maximizing this distance therefore favors satellites that add new coverage periods rather than simply increasing the number of satellites.

- Largest pass-time gap: The Wasserstein term rewards overall distributional difference, but does not specifically target the single largest empty stretch of the day. A dedicated, wraparound-aware function computes the largest gap, in hours, between the end of one pass and the start of the next in the combined constellation's daily schedule, treating the 24-hour day as circular, so a gap spanning midnight is correctly measured as one gap rather than being split in two at the day boundary. The reduction in this largest gap that a candidate achieves is added to the objective directly, and is also, on its own, enough to justify accepting a candidate explained in 3.1.2.
- Gap uniformity: A candidate that shrinks the single largest gap can still leave several other medium-sized gaps untouched. A gap-uniformity index, the sum of the squares of every circular gap in the daily schedule, each expressed as a fraction of the day is lower for an evenly spread schedule and higher when passes cluster with large gaps between clusters; the reduction a candidate achieves in this index is likewise added to the objective, so the search is rewarded for evening out the whole schedule, not only for patching its single worst hole.

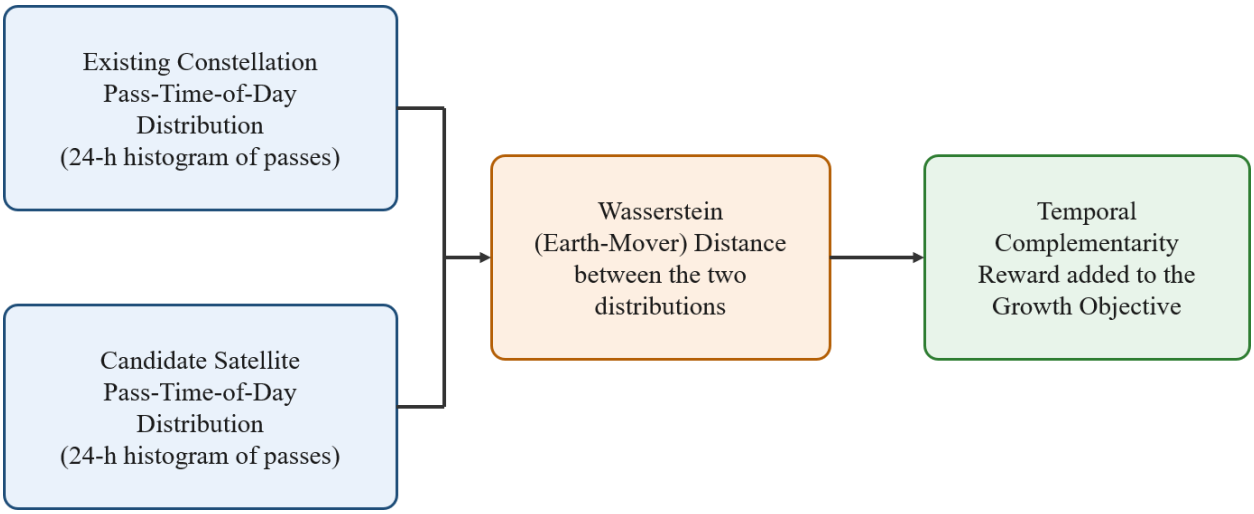


Figure 4: Wasserstein-Distance Temporal-Complementarity Term

Each of these three terms is normalized by its own scale and combined with a hand-set weight before being added to the mission score, in the same way the mission score's own sub-terms are combined. Together with the worst-revisit veto gives the growth loop a layered set of safeguards: a hard floor on per-lake revisit performance, and three complementary, increasingly specific

rewards: distributional complementarity, the single worst gap and overall evenness, for the shape of the daily schedule.

3.1.4 Coverage, Revisit and Ground-Station Metrics

Every candidate orbit, at every stage of the pipeline, is scored from the same underlying simulation output: the merged set of time intervals during which any satellite's sensor swath overlaps Nepal (the observation intervals) and the merged set of intervals during which any satellite is in contact with the ground station. From these, the pipeline derives: coverage fraction (the accumulated, de-duplicated swath area inside Nepal's border, divided by Nepal's area); average and maximum revisit time (the gaps between consecutive observation intervals); passes per day; total and average ground-station (SGS) contact time; average downlink latency (the time from an observation to the next ground-station contact); and, for each of six named glacial lakes (Imja Tsho, Tsho Rolpa, Thulagi, Chamlang South, Barun Tsho, and Lumding Tsho), the number of passes, dwell time, and whether it is covered at all. Coverage-area results are additionally cross-checked against an independent geometric reference: Nepal's border is also approximated as a 64-sided polygon, and the two area estimates are compared to quantify the numerical margin from that approximation.

3.1.5 Visualization and Verification Outputs

Three kinds of visual output are produced for every design: an interactive 3-D satellite scenario viewer showing every satellite's orbit, sensor footprint, and ground-station link; a static 2-D ground-track map of the whole constellation over Nepal, with each satellite drawn in its own consistent color; an animated version of the same 2-D map, stepped frame by frame through the 24-hour scenario; and a pass-time histogram, binned by hour of day and stacked per satellite using the same per-satellite color scheme as the ground-track maps, with the largest remaining gap shaded and labeled directly on the plot. Because the 3-D viewer's browser-based globe can occasionally time out when many satellites, sensors, and ground stations are being placed at once, playback is wrapped in a short retry loop that re-draws every sensor footprint and backs off between attempts, so a slow-to-initialize viewer does not stop the rest of the pipeline from running.

3.2 Remote Monitoring

Nepal has many remote, high-altitude and disaster-prone regions that are difficult to observe continuously from the ground. Beyond designing the orbits, the proposed LEO CubeSat constellation is intended to provide repeated satellite observations of such regions and reduce dependence on physical observation. The remote-monitoring application focuses on Glacial Lake Outburst Floods (GLOFs): observing glacier lakes such as Imja Tsho for changes in lake or water area.

SAR is proposed as the main remote-sensing payload. Optical sensors depend on sunlight and are blocked by clouds, whereas SAR uses radar signals. It can therefore observe during both day and night and is much less affected by cloud cover. This matters for Nepal, where mountainous regions often have cloudy and changing weather. SAR is also sensitive to water, which supports lake monitoring and repeated images allow multi-temporal change detection.

Imja Tsho was selected as the example target. The approach is to observe the lake with SAR on one date, observe it again on a later date, compare the two images and detect changes in the lake or surrounding water area. For this demonstration, Sentinel-1 SAR data were obtained and processed in Google Earth Engine (Figs. 1 and 2). Sentinel-1 is not the proposed CubeSat; its real SAR data are used as representative data to show the type of monitoring that the proposed CubeSat SAR payload could perform.

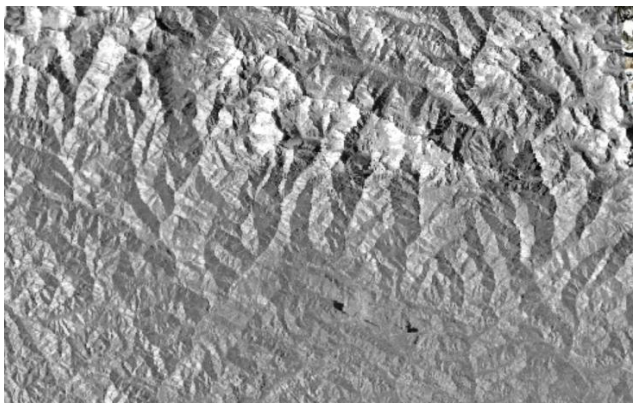


Figure 5: Sentinel-1 SAR image of mountainous terrain GEE



Figure 6: Sentinel-1 SAR image of the Imja Tsho region GEE

The satellite pass over a target starts the monitoring chain: CubeSat pass, SAR observation, transmission to a ground station, image processing, change detection, and finally GLOF. The constellation design determines how often this chain can run, so coverage, visibility, imaging

opportunity and revisit time over Nepal directly affect monitoring performance. The two parts of the project fit together as follows.

Part	Tool / data	Role in remote monitoring
Constellation analysis	MATLAB	Satellite position, ground track, visibility of a target such as Imja Tsho, imaging opportunities and observation times
SAR observation	Proposed CubeSat SAR payload	Acquires day-and-night, cloud-independent images of the target
Demonstration	Sentinel-1 SAR + Google Earth Engine	Representative real data used to show image processing and change detection

This work focuses on satellite-based observation and change detection. It does not directly predict a GLOF; it provides updated observations of glacier lakes and surrounding areas that can support further disaster-risk assessment. The demonstration uses Sentinel-1 data rather than data from the proposed CubeSat, so it illustrates the method rather than the performance of the proposed payload.The proposed LEO CubeSat constellation can provide an effective platform for remote monitoring of disaster-prone and inaccessible regions of Nepal. With a SAR payload, observations are possible by day and night and under cloud, which is particularly useful for GLOFs, floods and environmental change in mountainous regions. The Imja Tsho demonstration with Sentinel-1 data in Google Earth Engine shows how SAR images from different dates can be compared for change detection. Evaluating the constellation's coverage, visibility, imaging opportunities and revisit time then shows how frequently key regions such as Imja Tsho and flood-prone areas can be monitored. Remote monitoring thus links the constellation design, SAR payload, satellite observation and disaster-monitoring applications, demonstrating the potential usefulness of the proposed CubeSat system for Nepal.

3.4 Research Timeline

Table 1 shows the work timeline for the research.

Phase	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan
Literature Review										
Requirement analysis and mission planning										
Orbital and coverage modelling										
Image extraction for Remote Monitoring										
Software pipeline integration										
Result verification and optimization										
Project Demonstration										

Table 1: Research Timeline (Gantt Chart)

■ Work accomplished ■ Work to be completed

CHAPTER IV: RESULT AND ANALYSIS

The results below come from a representative, fully-logged design run of the pipeline described in methodology and are reported here with the exact figures that run produced. The constellation-growth objective's gap-uniformity and hourly-coverage terms were developed.

4.1 Simulation Results

4.1.1 Baseline Single-Satellite Orbit

The 100-sample Latin Hypercube seed plus 100-iteration Bayesian Optimization search converged on a best single-satellite orbit of inclination 28.66°, RAAN 345.45°, and true anomaly 32.74°, with an overall weighted mission score of 0.0347. Table 2 shows the best value the search found for each individual metric, and the orbit at which it occurred - useful for seeing how far a single-objective-best orbit would be from the multi-objective best overall orbit actually chosen.

Metric	Best value	Inclination (deg)	RAAN (deg)	True anomaly (deg)
Coverage fraction	0.4899	28.64	309.01	233.24
Passes per day	3.0000	28.23	20.31	180.24
SGS total contact (min)	80.0000	28.57	251.99	54.68
Average revisit (min)	100.0000	28.88	178.85	235.77
Average latency (min)	93.0000	89.08	311.48	268.90
Lakes covered	6.0000	28.63	232.05	31.19
Worst-case lake revisit (min)	Inf	27.43	295.95	226.34
Min. passes/day over any lake	1.0000	28.63	232.05	31.19

Table 2: Best-Per-Metric Results from the Baseline BO Search

Re-simulated at high resolution, the chosen baseline orbit achieves an average revisit time of 99.47 minutes (maximum 99.63 minutes) over 3 passes per day, 77.33 minutes of total ground-station contact across 7 contacts, a coverage fraction of 0.8511, and an average downlink latency of 94.33 minutes. Its three daily passes cover 1,595.22 km², 71,885.65 km², and 89,418.83 km² respectively, for a de-duplicated (union) swath area of 125,578.14 km² inside Nepal. This union area is cross-checked against an independent geometric estimate - Nepal's border approximated as a 64-sided polygon, giving a reference area of 125,913.23 km², confirming that the swath-accumulation calculation is consistent.

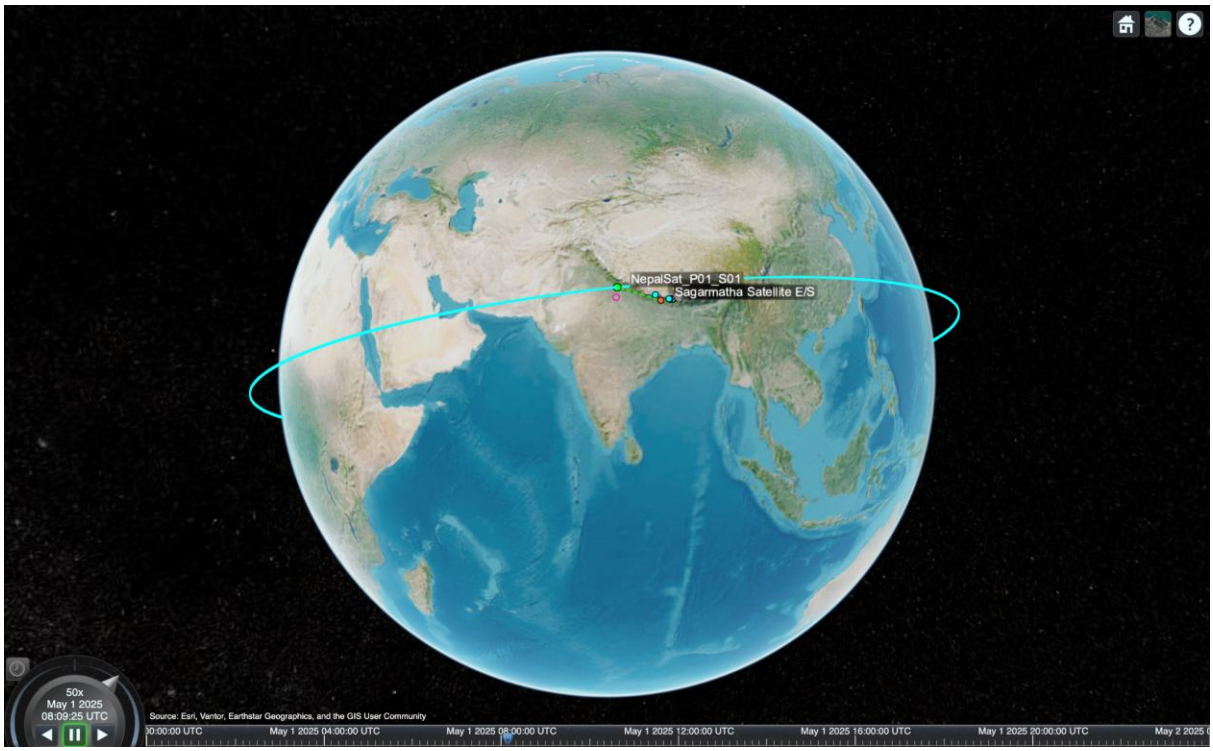


Figure 7: Best single-satellite orbit with sensor footprint and GS link

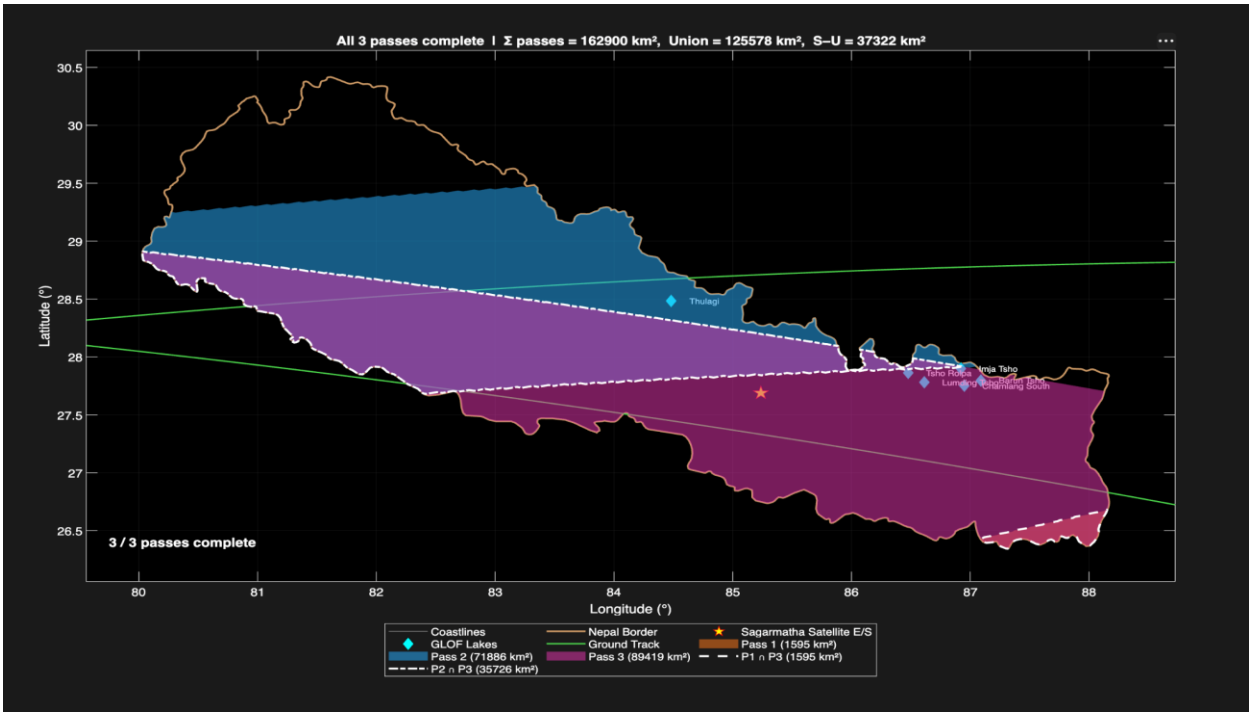


Figure 8: 2-D ground-track map over Nepal

4.1.2 Incremental Constellation Growth

From this baseline, the growth loop ran for 6 steps within its budget of 3 orbital planes and 3 satellites per plane, accepting every step it proposed. Table 3 reproduces the full step-wise log: two satellites were added to the baseline's orbital plane, a second plane was then found and filled with two more satellites, and a third plane was added with its first satellite, for a final constellation of 7 satellites across 3 orbital planes.

Step	Action	Incl. (deg)	RAAN (deg)	v (deg)	Alt. offset (km)	Combined coverage	Combined revisit (min)	Wasserstein
1	Add Satellite	28.662	345.45	215.12	-15.0	66.28%	49.50	0.070
2	Add Satellite	28.662	345.45	332.35	+15.0	78.34%	30.71	0.044
3	Add Orbit	32.157	166.47	108.56	-35.1	85.91%	103.22	1.012
4	Add Satellite	32.157	166.47	28.84	+8.4	86.71%	80.42	0.765
5	Add Satellite	32.157	166.47	312.69	-20.1	86.74%	65.07	0.607
6	Add Orbit	31.972	0.00	127.49	+34.6	92.56%	57.41	0.424

Table 3: Incremental Constellation Growth

The accepted reasons recorded against each step show the growth objective working as intended: step 1 was accepted for a +19.9% coverage gain, a 51.0-minute revisit reduction, and a 51.5-minute latency reduction; step 3 (the first new orbital plane) for a +7.6% coverage gain, a 72.5-minute revisit reduction, and the largest single-step Wasserstein score (1.01) of the run, reflecting how different that new plane's pass times were from the existing one-plane schedule; and step 6 (the third plane) for a +5.8% coverage gain even though its own revisit contribution was a 7.7-minute regression, illustrating that the growth objective's other terms can justify a step the revisit-time metric alone would not.

The final, 7-satellite, 3-plane constellation reaches an average revisit time of 57.41 minutes (maximum 391.00 minutes), 18 passes per day, 523.00 minutes of total ground-station contact across 49 contacts, 92.56% coverage of Nepal, and an average downlink latency of 25.94 minutes - a clear improvement over the single-satellite baseline on every metric except maximum revisit

time. Accumulated swath coverage over the 24-hour scenario reaches 136,569.63 km² from the growth loop's own (60-second-sample) metrics, and 142,022.39 km² of union swath from the separately rendered, higher-resolution static and animated ground-track maps because of the resolution-dependent gap.

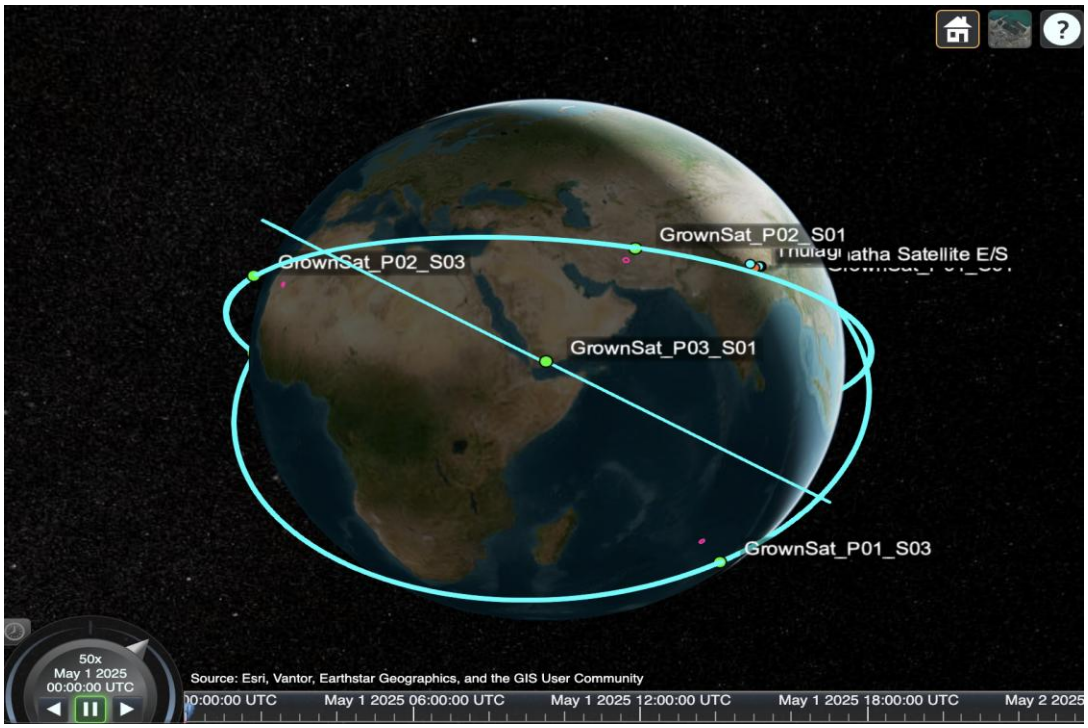


Figure 9: Final grown constellation

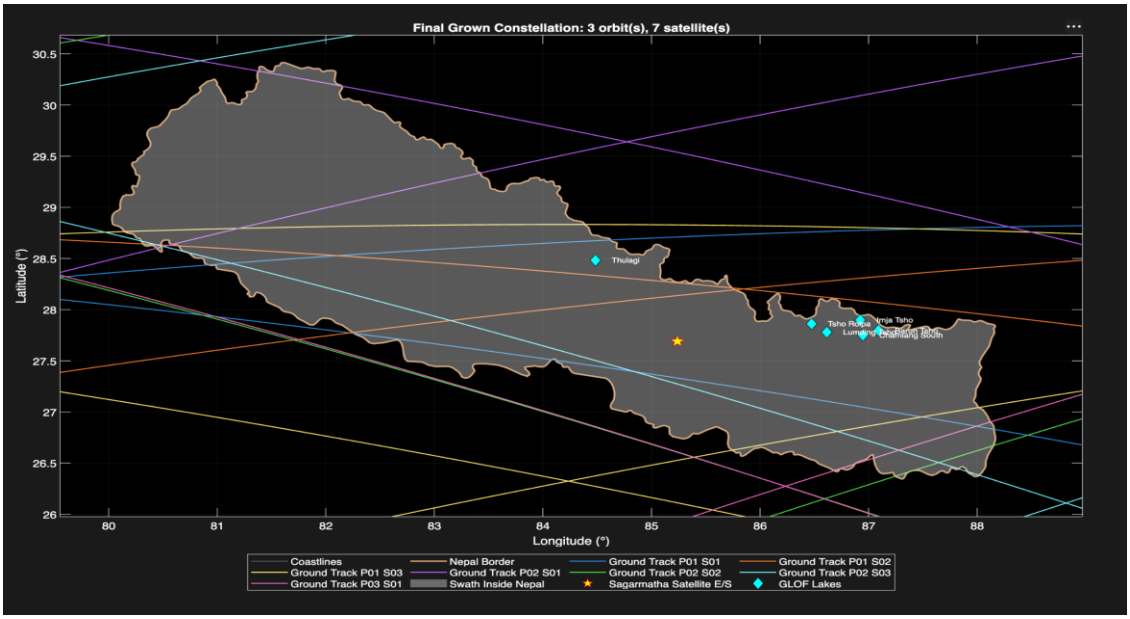


Figure 10: Final grown constellation , static 2-D ground-track map

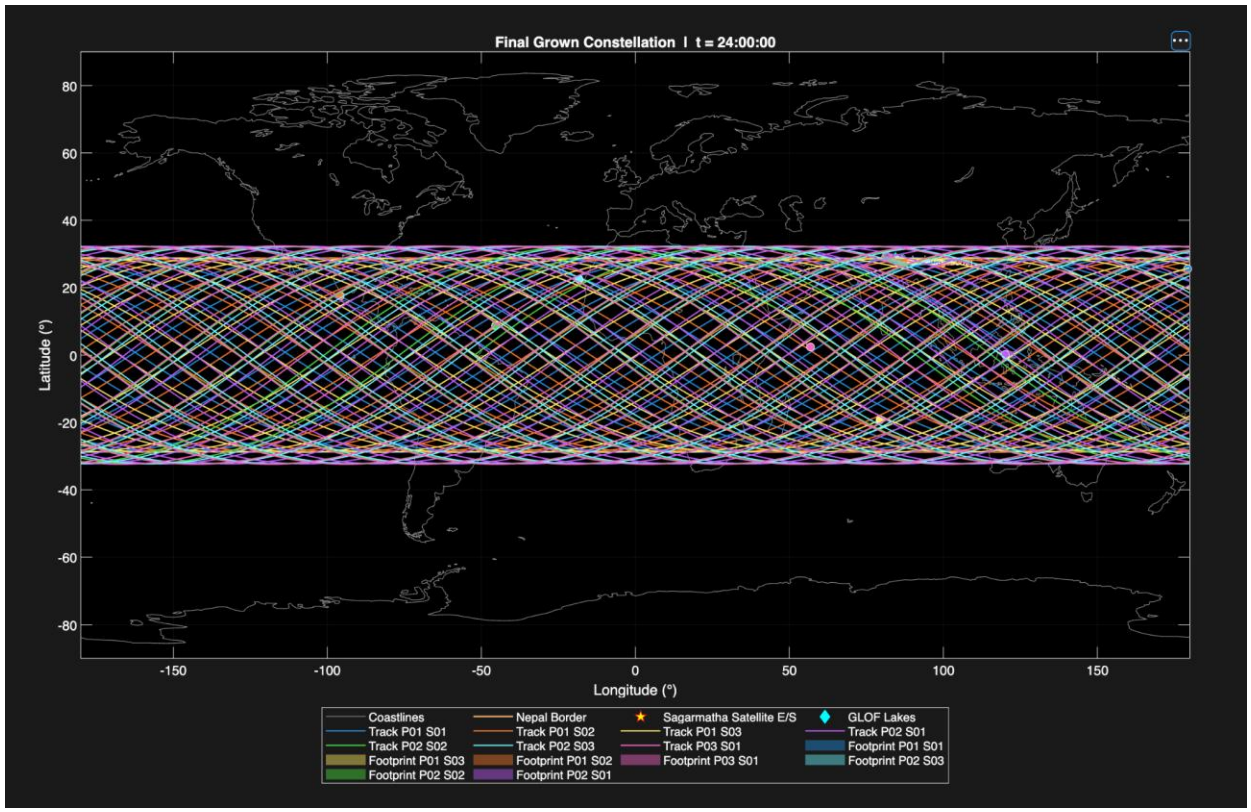


Figure 11: Animated 2-D ground-track build-up

4.1.3 Per-Glacial-Lake Coverage

Table 4 shows per-lake coverage for the baseline single-satellite orbit: all six named glacial lakes are covered at least once per day, each by exactly one pass. For the final, 7-satellite constellation, the pipeline reports the lakes still fully covered (6 of 6) with a worst-case lake revisit time of 48.00 minutes, confirming that growth did not sacrifice lake coverage even as it was optimized primarily for country-wide metrics; a full re-tabulated per-lake breakdown for the grown constellation

Lake	Passes	Mean revisit	Max revisit	Dwell (min)	Covered
Imja Tsho	1	N/A	N/A	0.10	true
Tsho Rolpa	1	N/A	N/A	0.23	true
Thulagi	1	N/A	N/A	0.43	true
Chamlang South	1	N/A	N/A	0.27	true
Barun Tsho	1	N/A	N/A	0.20	true
Lumding Tsho	1	N/A	N/A	0.27	true

Table 4: Per-Glacial-Lake Coverage Summary by Single-Satellite

Mean and maximum revisit are reported as N/A because, with exactly one pass per lake per day in the baseline orbit, there is only one gap to measure (between that pass and its own recurrence the next day), which the metric does not treat as a same-day revisit statistic. Lakes covered: 6/6; minimum passes per day over any lake: 1; mean passes per day: 1.00.

```
========= FINAL GROWN CONSTELLATION METRICS =========
Revisit time with swath:
  Average: 57.24 min
  Maximum: 215.00 min
Number of passes per day within swath: 18
SGS contact:
  Total contact time: 510.00 min
  Average per contact: 12.14 min
  Number of contacts: 42
Swath overlap time:
  Total: 10.00 min (41.67% of day)
Coverage fraction (accumulated area / Nepal area): 0.9625
Data latency (average time from observation to next SGS contact):
  Average: 35.56 min
Accumulated coverage over 24 h: 142022.39 km^2
Six lakes covered: 6/6; worst lake revisit: 31.00 min
Largest daily pass-time gap: 7.62 hr (baseline was 20.62 hr)
=====================================================
```

Figure 12: Final Grown Constellation Metrics

4.1.4 Temporal Gap and Hourly-Coverage Analysis

Plotting the final constellation's passes by local hour of day as shown in Figure 11, shows that the 18 daily passes are not spread evenly across the 24-hour day: they cluster into three windows - roughly 06:00-11:00 and 13:00-16:00 and 17:00-23:00 - contributed respectively by the three orbital planes added in Table 3, leaving blocks of the day, most persistently 00:00-05:00, with no pass at all. The largest single gap measured in one such run was 7.62 hours, spanning from the last pass of one day to the first of the next.

The cause is geometric rather than a search failure: every orbital plane found by the growth loop has an inclination of roughly 28.7-32°, close to Nepal's own latitude band (about 26-30°N), so each satellite passes over Nepal near the northern apex of its ground track, where the track runs nearly parallel to Nepal's latitude. Earth's rotation brings that apex over Nepal's longitude only once per day, so a single orbital plane contributes one pass-time window a few hours wide, not

passes spread through the day. Satellites added to the same plane can only shift a pass within or adjacent to that plane's own window by changing true anomaly and altitude offset; only a new orbital plane, with its own RAAN, can reach a genuinely different window. With the growth budget fixed at 3 planes, the daily schedule is structurally limited to about 3 windows, and some gap between them is close to unavoidable at this budget.

This finding directly motivated the largest-gap and gap-uniformity terms in the growth objective : earlier in development, the Wasserstein-distance term alone was found to reward overall distributional spread without specifically targeting the one largest empty stretch, so it narrowed the gaps only weakly. Giving the largest-gap term too large a weight relative to the mission score was also found to be counter-productive on its own: because same-plane growth cannot reach a new time-of-day window, an over-weighted gap term pushed the same-plane (inner) search to chase an unreachable target and sacrifice coverage and revisit performance for little or no gap reduction, while the 3-plane ceiling left the true cause - too few orbital planes, not poorly chosen altitude offsets - untouched. The current objective instead balances the Wasserstein, largest-gap, and gap-uniformity terms as softer, co-equal rewards evaluated against the existing 3-plane/7-satellite budget, with raising that budget - and explicitly spacing new planes' RAAN around the day, rather than placing them greedily - identified as the more direct fix.

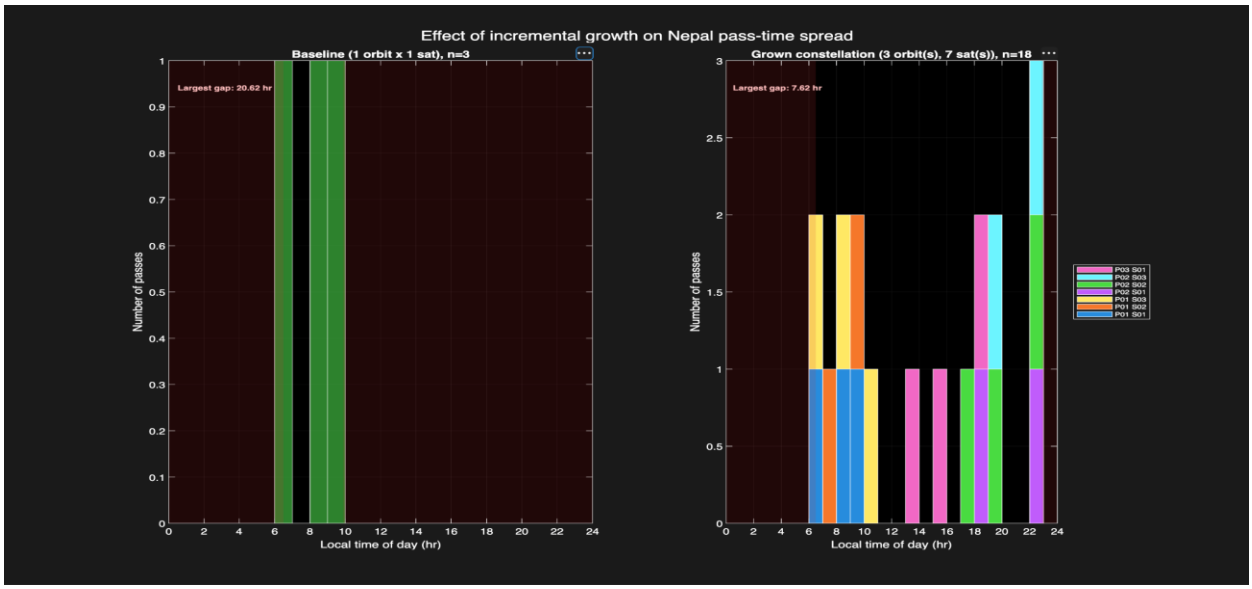


Figure 13: Pass-time-of-day histogram: single satellite vs. final grown constellation

4.2 Works Accomplished

- Formulated single-satellite orbit selection as a Bayesian-Optimization problem (Latin Hypercube seeding, Gaussian-Process surrogate, Expected-Improvement acquisition) over inclination, RAAN, and true anomaly, scored against a weighted, multi-metric mission score.
- Built an incremental constellation-growth loop that alternates inner (same-plane) and outer (new-plane) Bayesian-Optimization searches, subject to a hard worst-case-lake-revisit veto, and grew a single satellite into a 7-satellite, 3-orbital-plane constellation.
- Extended the growth objective beyond the mission score with a Wasserstein-distance temporal-complementarity term, and with explicit largest-gap and gap-uniformity terms that directly target a narrow, evenly spread daily revisit schedule.
- Implemented the full coverage/revisit/ground-station/latency/per-lake metrics pipeline, and cross-checked its swath-area calculation against an independent 64-sided-polygon geometric reference.
- Produced static and animated 2-D ground-track maps and a 3-D satellite scenario viewer, each with consistent per-satellite coloring, and a pass-time histogram with an explicit largest-gap annotation; hardened the 3-D viewer's playback against intermittent initialization timeouts.
- Diagnosed, from the resulting pass-time histogram, the geometric reason the current 3-plane budget leaves a persistent gap in the daily schedule, and used that diagnosis to shape the growth objective's gap-related terms.

4.3 Works Remaining

Future iterations of this project will focus on the following tasks:

- To re-run the pipeline with the current gap-uniformity-weighted growth objective and report an updated set of results, including a full per-lake breakdown for the grown constellation.
- To close the residual daily coverage gap, either by raising the orbital-plane budget beyond 3, by exploring higher-inclination auxiliary planes or by spacing new planes' RAAN explicitly rather than placing them greedily.
- To extend the simulation from a single 24-hour scenario to multi-day and end-of-life analysis, incorporating J2 and atmospheric-drag perturbations, orbital decay, and a station-

keeping/reboost schedule, so that the gap and uniformity results reported here can be shown to hold (or be shown to degrade, and by how much) over the mission's lifetime.

- To complete Remote Monitoring: the ground-station uplink/downlink scheduling and onboard sensing workflow that turns a pass over a hazard zone into usable, downlinked imagery.
- To integrate the features of MATLAB related to orbital design and Google Earth Engine related to remote monitoring.
- To validate the design against representative Two-Line Element (TLE) data for an existing constellation of similar altitude and inclination, as an independent check on the SGP4-propagated results reported here.
- To detail coverage, revisit time and latency of targeted GLOF sites.

4.4 Problems Encountered

- The browser-based 3-D viewer can intermittently time out while initializing large scenarios, so a retry-and-backoff wrapper was added to re-issue sensor fields of view before each retry, although this remains a limitation of the viewer.
- The standard revisit-time metric checks only gaps between passes within a day and misses the wraparound gap between the last pass of one day and the first of the next, so a dedicated wraparound-aware largest-gap function was added.
- Heavily weighting the largest-gap term initially worsened results because the same-plane search could not reach the required different time of day, which required a new orbital plane, so the growth sequence was analyzed before adjusting the weights.
- Each Bayesian Optimization candidate requires full orbit propagation and coverage simulation, making new-plane searches expensive and limiting how thoroughly inclination and RAAN can be explored, which is why the current growth budget is capped at 3 planes and 6 steps.

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